

Year-round operational optimization of HVAC systems using hierarchical deep reinforcement learning for enhancing indoor air quality and reducing energy consumption

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HIGHLIGHTS

- Designed a hierarchical deep reinforcement learning (HDRL) control framework to address challenges in training with hybrid action spaces.
- Developed a simulation platform integrating heating, ventilation and air conditioning (HVAC) system and building energy models for reliable agent training and performance evaluation.
- Developed a comprehensive multi-objective reward function to balance indoor temperature, air quality, and energy efficiency.
- Achieved year-round operational performance improvements, broadening the applicability of DRL in HVAC control.

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ABSTRACT

Balancing user comfort demands with energy consumption in heating, ventilation, and air conditioning (HVAC) systems is a critical challenge in the field of control optimization. Currently, deep reinforcement learning (DRL) algorithms have achieved some success in optimizing HVAC system control. However, training agents with existing algorithms becomes challenging, when dealing with the hybrid action control problem. This study proposes an HVAC system optimization control method based on hierarchical deep reinforcement learning (HDRL), which aims to minimize system energy consumption while ensuring indoor comfort and maintaining air quality. The proposed HDRL method employs a two-layer agent architecture, where the bottom agent is responsible for optimizing the frequency control of fans and pumps and the top agent flexibly selects the daily operating mode based on outdoor meteorological conditions. We designed a multi-objective reward function based on Gaussian distribution, hyperbolic tangent function, energy consumption normalization for bottom agents, and temperature-based reward function for the top agent to achieve the reliable training of the agents. Experimental results indicated that the HDRL method significantly improves indoor temperature and CO₂ concentration control during the year-round operation of the HVAC system. Compared to conventional PI control, the temperature compliance rate is increased by 8.37 %, and the combined temperature and CO₂ compliance rate is improved by 7.8 %, while achieving a 3.95 % reduction in energy consumption.

1. Introduction

1.1. Background

With the rapid development of the socio-economic landscape, human activities have increasingly shifted indoors, where individuals now spend 80 % to 90 % of their time [1]. This shift has raised concerns about indoor thermal comfort and air quality, especially with the emergence

of sick building syndrome (SBS) [2,3]. Heating, ventilation and air conditioning (HVAC) systems are essential for maintaining indoor comfort but account for nearly 50 % of building energy consumption [4]. Thus, optimizing HVAC systems to reduce energy use while maintaining occupant comfort has become a major focus for researchers and practitioners [5,6].

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Existing studies have proposed various optimal control strategies to balance comfort and energy consumption, including rule-based, model-based, and model-free approaches [7].

Rule-based control, such as Proportional–Integral–Derivative (PID) control, is commonly used in HVAC systems. PID relies on a feedback loop to adjust control variables (CV) based on the error between the process variable (PV) and the setpoint (SP) [8]. However, fixed rules of PID often lack flexibility in adapting to dynamic, nonlinear environments, and tuning PID gains can be complex and time-consuming, leading to performance degradation under varying conditions [9]. Model-based control is considered an effective approach to address these challenges, by optimizing control variables using predictive models and algorithms [10,11]. Model Predictive Control (MPC), an advanced model-based control method, has demonstrated significant potential in energy savings [12–14]. However, its effectiveness depends on model accuracy and is constrained by computational speed and real-time feasibility [15,16].

In recent years, with advances in big data and computing power, Reinforcement Learning (RL) has been widely explored and applied. RL agents learn optimal control policies by interacting with the environment, making RL particularly suitable for sequential decision-making problems involving environmental uncertainties [17], and therefore, it has gained considerable attention in HVAC system control optimization. Typical RL algorithms are model-free, and compared to model-based approaches, RL agents can be efficiently deployed after training, which reduces computational costs in real-time optimization.

1.2. Literature review for deep reinforcement learning control

Traditional RL methods utilize tables to record states and actions, making it challenging for agents to learn effectively when faced with complex policies. In real-world HVAC system control, continuous and high-dimensional states and actions are more common [18]. Compared to RL, Deep Reinforcement Learning (DRL) integrates Deep Neural Networks (DNNs) with RL algorithms, enabling the learning of effective policies in high-dimensional and complex state spaces [19].

Thus far, numerous scholars have applied DRL to HVAC system control to balance the trade-off between occupant comfort and energy consumption. Nguyen et al. [20] employed the Phasic Policy Gradient (PPG) algorithm to optimize HVAC systems, enhancing energy efficiency while maintaining indoor temperature comfort. Gao et al. [21] innovatively introduced predictive information into DRL algorithms, proposed an optimization approach combining the Gated Recurrent Unit (GRU) and Soft Actor-Critic (SAC), and validated its effectiveness in optimizing HVAC operational strategies on the BOPEST platform. Zhang et al. [18] proposed a DRL framework for building energy system control, integrating the asynchronous Advantage Actor-Critic (A3C) algorithm to control a hydronic radiant heating system, which reduced heating demand by 16.7 % during winter deployment. However, this study did not address air quality control, which is a limitation that is also present in [22–25]. Cui and Xue [26] proposed a preference-informed TD3 algorithm, which comprehensively considers comfort, air quality, and system efficiency. An and Chen [27] used the Deep Q-learning (DQN) algorithm to control air conditioning, air purifiers, and window operation, demonstrating the potential of DRL in air quality control. Valladares et al. [28] utilized a Double Q-learning algorithm to optimize air conditioning and ventilation control in classrooms, improving both comfort and air quality, with validation conducted in real environments. However, the air conditioning systems involved were relatively simple. Although these studies have achieved progress in air quality control, further research is needed to address complex HVAC systems and long-term operational effects. Table 1 summarizes the application studies of DRL in HVAC control.

Through comprehensive review and discussion, we identified that despite DRL methods having made significant achievements in the application of HVAC control optimization, several critical challenges still exist:

- Existing methods rarely address HVAC control in complex hybrid action spaces, with most focusing on either discrete or continuous action spaces. However, real-world HVAC systems require hybrid control; for example, during working hours, it is necessary to simultaneously consider the on/off state of the heat pump (discrete values), number of units turned on (discrete values), and pump frequency (continuous values). Failure to address this issue limits the applicability of DRL in the overall system.
- Most studies focus on energy savings or thermal comfort, with fewer considering CO₂ concentration control, and even fewer addressing all three objectives simultaneously. Multi-objective control increases demands on reward function design, underscoring the importance of robust reward function development.
- Many DRL control studies focus on optimizing HVAC performance for specific seasons or time periods; however, real-world HVAC systems, often require stable operation throughout the year. Although DRL agents can be trained separately for different seasons or requirements and manually switched, such mode switching is suboptimal. Therefore, a DRL control strategy suitable for year-round operation is required.

1.3. Contribution of this study

In response to these challenges, a hierarchical RL agent control framework is proposed to optimize HVAC system year-round operation. The framework aims to enhance indoor environmental quality while improving energy efficiency. It is designed to flexibly adapt to various energy usage scenarios throughout the year-round operation of HVAC systems. The main contributions of this study are summarized as follows:

- A hierarchical deep reinforcement learning (HDRL) method is proposed for year-round HVAC system optimization. Utilizing TD3 and PPO algorithms, the hierarchical training framework addresses the challenges of hybrid action spaces, which are a common issue in HVAC system automation. This approach represents a key advancement for applying DRL to HVAC optimization.
- A robust reward function is developed that simultaneously considers indoor temperature control, air quality control, and system energy efficiency. To the best of our knowledge, this is the first study for utilizing a hyperbolic tangent function-based CO₂ concentration control reward function to enable effective air quality control by the agent.
- The proposed algorithm is applicable for year-round HVAC system operation, a wide range of time frames and diverse operating modes, providing strong reference value for practical applications.

The remainder of this paper is organized as follows: Section 2 provides a detailed introduction to the basic concepts of the DRL algorithms involved and the proposed HDRL framework. Section 3 describes the modeling process of the HVAC system environment in this study. Section 4 presents the details of the case study. Section 5 discusses the results of the case study. Finally, Section 6 concludes the paper and provides future perspectives.

2. Methodology

2.1. Basic definitions and methods of reinforcement learning (RL)

The RL agents learn environment features through interaction, which adjusts their behavior based on rewards to maximize cumulative returns. Typically, RL control problems are modeled as a Markov Decision Process (MDP), described by a quintuple (S, A, P, R, γ) . For the HVAC system control problem, the MDP components are:

- State space S : System states observable by the agent, $S = \{s_1, s_2, \dots, s_n\}$. For HVAC control, these states include indoor environmental parameters (e.g., temperature, humidity, and air quality), equipment states (e.g., air conditioning mode, and air flow rate), and outdoor conditions (e.g., temperature, humidity, and solar radiation).

Table 1
Summary of DRL control in HVAC system.

Ref.	Control variables	Action space type	Control objectives	Algorithm	Validation period
[20]	Indoor temperature setpoint	Continuous	Indoor temperature, energy consumption	PPG	Not specified
[21]	Supply air temperature setpoint, damper position	Continuous	Comfort, system operating cost	SAC	Summer and winter; 1 week
[18]	Supply water temperature setpoint, HVAC status (on/off)	Discrete	Comfort, energy consumption	A3C	Winter; 3 months
[22]	Damper position, window opening, fan speed	Continuous	Comfort, energy consumption	DDPG	Summer; 3 months
[23]	Supply water temperature setpoint	Discrete	Indoor temperature, energy consumption	DDQN	Winter; 3 months
[24]	Indoor temperature setpoint, air flow rate	Continuous	Indoor temperature, energy consumption	SAC, TD3, TRPO, PPO	All year
[25]	Supply air temperature setpoint, supply water temperature setpoint	Discrete	Indoor temperature, energy consumption	DQN	Summer; 2 months
[29]	HVAC output power	Discrete	Indoor temperature, energy consumption	DQN	Winter; 2 days
[30]	Indoor temperature setpoint	Discrete	Indoor temperature, energy consumption	DQN	Not specified
[31]	HVAC output power	Continuous	Comfort, system operating cost	DDPG, DDQN, SAC	Summer and winter; 2 days
[32]	Indoor temperature setpoint, fan mode	Discrete	Comfort, energy consumption	BDQ	Summer; 2 weeks
[33]	Supply water temperature setpoint	Discrete	Comfort, energy consumption	Q-learning	Summer; 2 months
[34]	Air flow rate change	Discrete	Airflow, CO ₂ concentration, energy consumption	SAC	1 day
[35]	VAV status (on/off)	Discrete	Indoor temperature, energy consumption	DQN	All year
[36]	Window status (on/off), HVAC status (on/off), temperature setpoint	Discrete	Indoor temperature, energy consumption	DQN	1 week
[37]	Cooling tower fan frequency, pump frequency	Discrete	COP	Q-learning	Summer; 3 months
[26]	Air flow rate change and fresh air ratio change	Continuous	Indoor temperature, CO ₂ concentration, energy consumption	TD3	Summer; 1 month
[27]	Window status (on/off), HVAC status (on/off), air purifier status (on/off)	Discrete	Indoor temperature, indoor PM2.5 concentration, energy consumption	DQN	Several days
[28]	Indoor temperature setpoint, fan status (on/off)	Discrete	PMV, CO ₂ concentration, energy consumption	Double Q-learning	Summer; 6 months

- Action space A : Actions available to the agent in each state, $A = \{a_1, a_2, \dots, a_n\}$. For HVAC systems, the actions include adjusting temperature setpoints, regulating frequency, and controlling on/off status.
- State transition probability P : Transition probability $p(s'|s, a)$ denotes the likelihood of moving to state s' after action a in state s . In HVAC control, this probability is often unknown, which makes model-free RL necessary for estimating transitions using historical data.
- Reward function R : Immediate reward $r = R(s, a)$ represents the reward for action a in state s . For HVAC systems, the reward function typically reflects energy cost, air quality, and thermal comfort, evaluating the actions of the agent.
- Discount factor γ : This factor quantifies the importance of future rewards, which ranges from 0 to 1. For HVAC systems, a smaller γ

emphasizes immediate control performance, while a larger γ focuses on future impacts.

The goal of the agent is to learn a policy $\pi(a|s)$ that maximizes the expected cumulative reward R_t , defined as:

$$R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}. \quad (1)$$

The MDP provides the standard framework for RL.

2.2. Deep reinforcement learning

DRL algorithms can be broadly categorized into three main types: value-based methods, policy-based methods, and Actor-Critic (AC) methods.

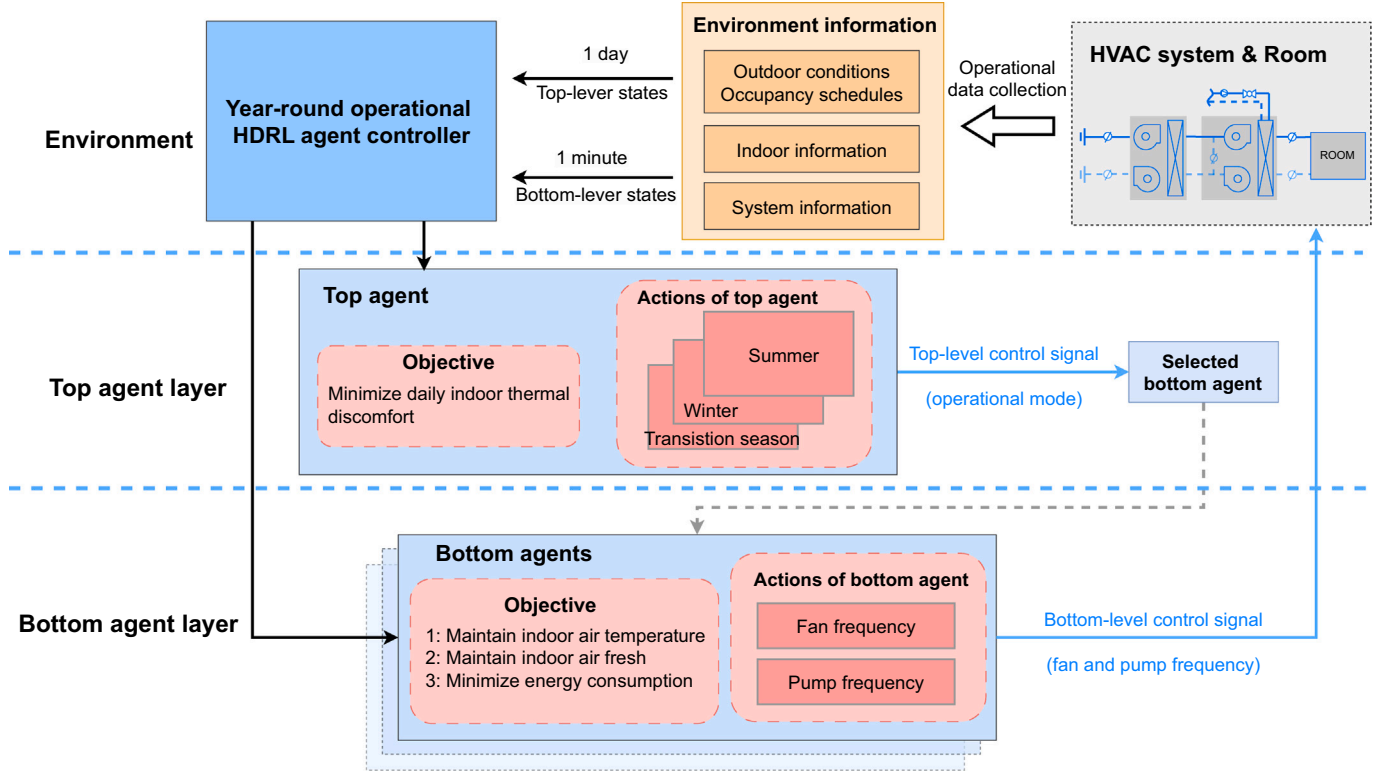


Fig. 1. HDRL control framework for optimizing HVAC system.

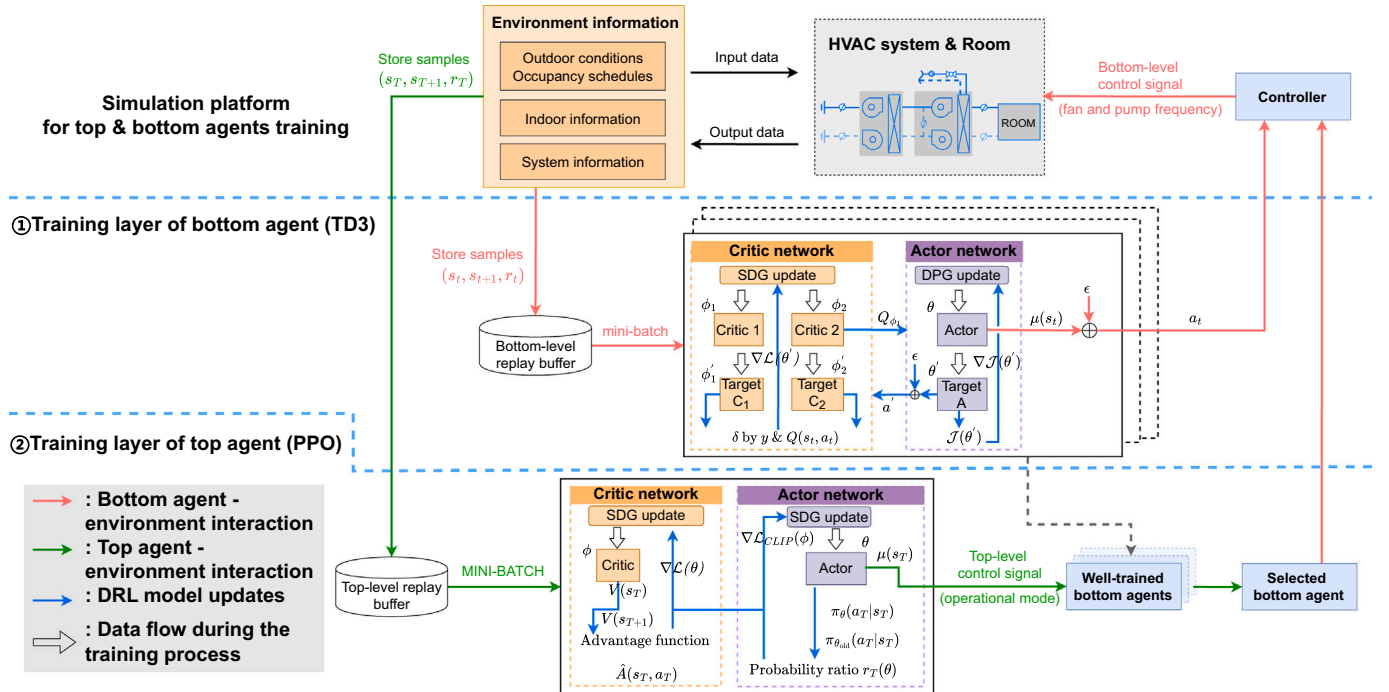


Fig. 2. Training process of the HDRL algorithm for HVAC system control.

2.2.1. Actor-Critic method

The Actor-Critic (AC) method combines the advantages of value-based and policy-based approaches, where a Critic network evaluates the action-value function $Q(s, a)$, and an Actor network selects actions a to maximize cumulative reward R_t . The Critic minimizes the

Temporal Difference (TD) error, providing critical feedback for updating the Actor's policy through policy gradients.

The policy gradient for the Actor is defined as:

$$\nabla_{\theta} J(\theta) = \mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t; \theta) \cdot \delta], \quad (2)$$

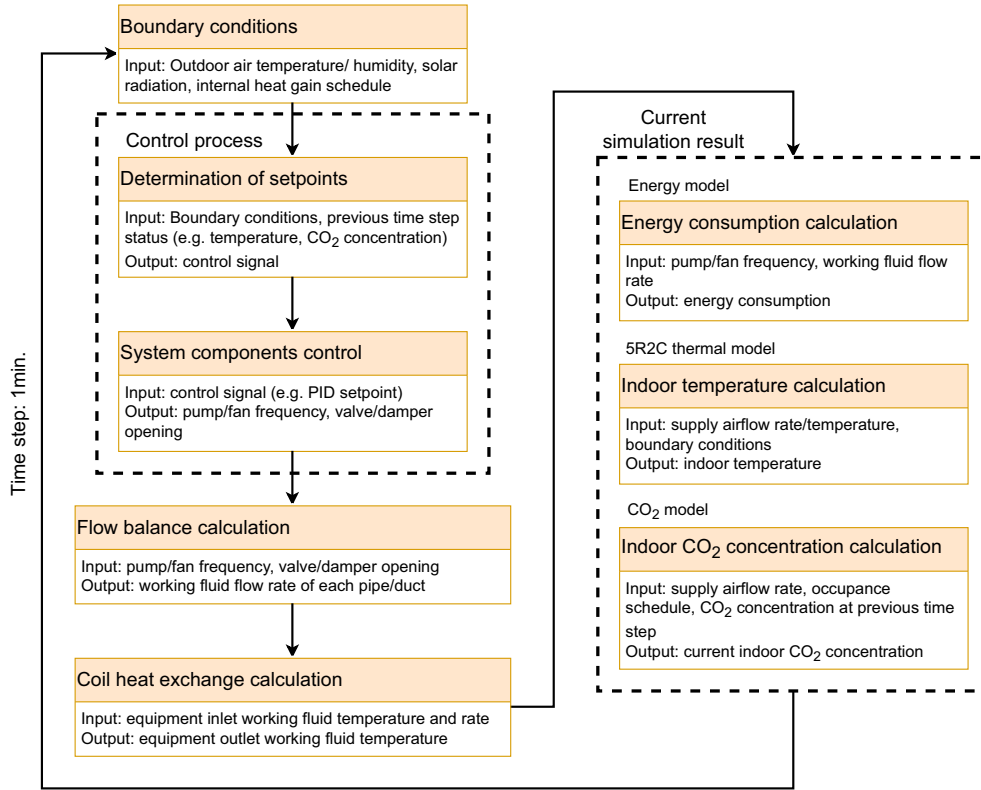


Fig. 3. Flowchart of the simulation program.

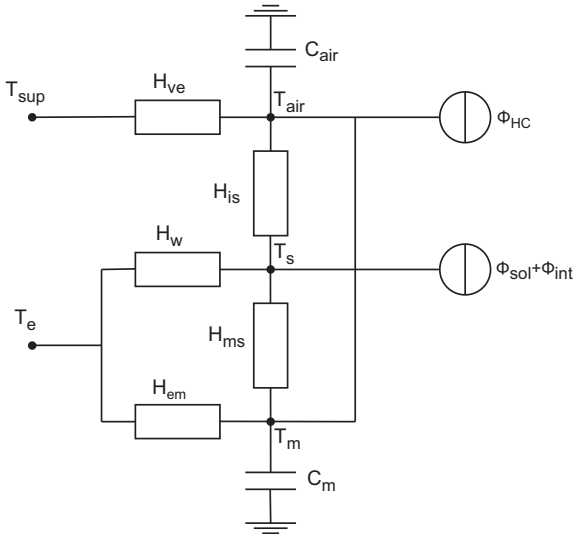


Fig. 4. RC thermal network (five-resistance, two-capacitor model (5R2C)).

Table 2

Summary of parameters for the 5R2C model.

Parameter [Unit]	Value
Window area [m ²]	35.2
Wall area [m ²]	336.8
Floor area [m ²]	160.0
Room Volume [m ³]	940.0
u walls [W/(m ² K)]	1.015
u windows [W/(m ² K)]	6.0
Room capacitance [J/K]	12,800,000 [42]

2.2.2. Twin Delayed Deep Deterministic Policy Gradient (TD3)

TD3 [38] is an enhanced version of the Deep Deterministic Policy Gradient (DDPG) algorithm [39], designed to address the overestimation bias in value function approximation, which is a common challenge in the AC methods. TD3 incorporates several key modifications to enhance learning stability, including the use of twin Critic networks, delayed updates for the Actor, and target policy smoothing. These features make TD3 well-suited for tasks involving continuous action spaces, such as controlling the fan and pump frequencies in the HVAC system.

2.2.3. Proximal Policy Optimization (PPO)

PPO [40] is a policy gradient method that provides a balance between stability and computational efficiency. PPO utilizes a clipped surrogate objective function to limit the magnitude of policy updates, thereby preventing the policy from changing too drastically during each iteration, which helps manage the policy changes effectively. In addition, sample efficiency and ease of implementation of the PPO further make it a preferred choice for discrete decision making in our proposed control framework.

where δ represents the TD error guiding updates. The Actor network is updated via gradient ascent, while Critic network minimizes δ through gradient descent.

In this study, we focus on two specific AC methods: Twin Delayed Deep Deterministic Policy Gradient (TD3) and Proximal Policy Optimization (PPO). TD3 is employed for continuous control, while PPO is responsible for discrete control. These algorithms are selected because of their strengths in addressing different types of action spaces and maintaining stable learning. Further details on AC network updates, TD3, and PPO algorithms are presented in [Appendix A](#).

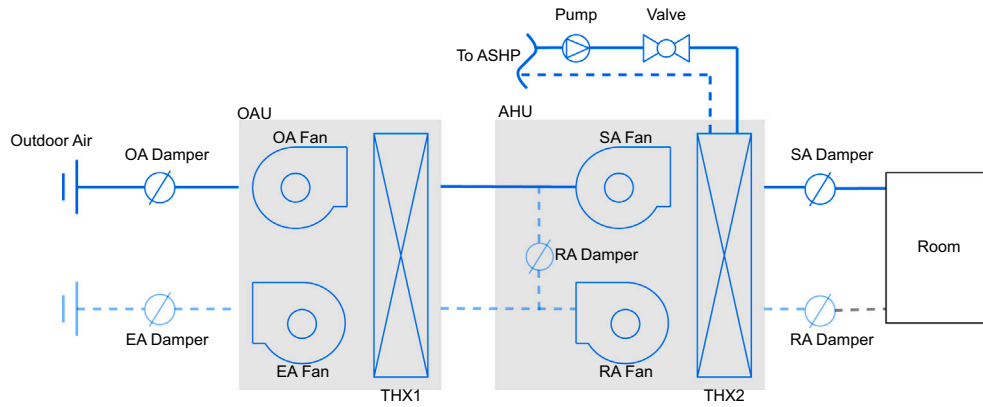


Fig. 5. Conceptual diagram of the system.

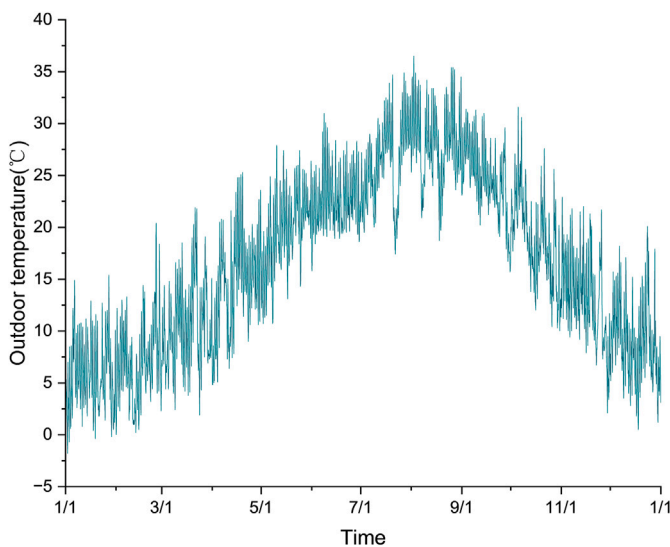


Fig. 6. Outdoor air temperature.

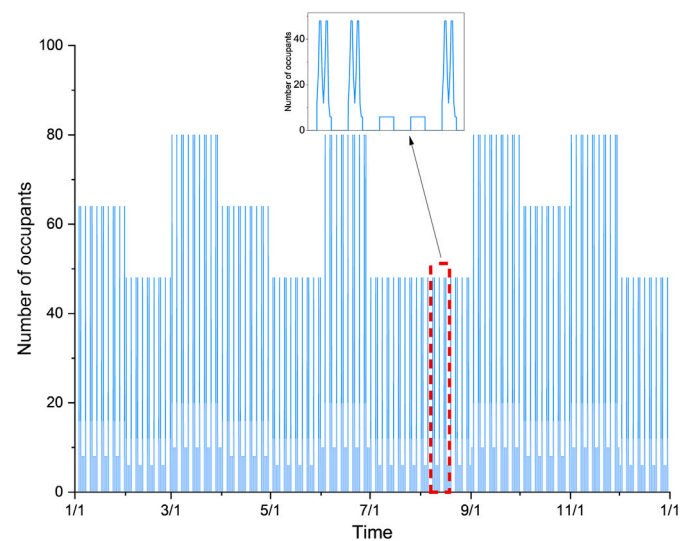


Fig. 7. Occupancy schedule in large lecture theater.

2.3. Hierarchical deep reinforcement learning proposed in this study

HDRL is a DRL method designed to decompose complex tasks through a hierarchical structure, which enables high-level and low-level policies to collaborate. The key components of HDRL are as follows:

- **Hierarchical policies:** High-level policies focus on global decisions, such as subtask selection, which often operate on discrete action spaces at lower frequencies (e.g., every N steps). Low-level policies execute detailed actions to implement high-level decisions, which operate on continuous action spaces at each step. This distinction in objectives and temporal granularity ensures effective task decomposition.
- **Hierarchical rewards:** The rewards are distributed based on the hierarchical structure. High-level rewards are sparse, targeting global objectives and long-term planning, while low-level rewards are dense, thereby focusing on local execution quality.

HDRL can be implemented via independent or joint training. Independent training sequentially trains low-level policies for subtasks and then high-level policies for invoking them, which offers simplicity and modularity but potentially less flexible policy collaboration. Joint

training simultaneously trains high- and low-level agents, thereby capturing dynamic dependencies but increasing complexity and slowing convergence.

2.4. Proposed method for HVAC system control

In classical AC algorithms, the Actor network learns either the probability distribution of discrete actions or the distribution parameters of continuous actions to optimize policies. However, when faced with hybrid action spaces, the dimensional and definitional differences between discrete and continuous actions make it difficult for the Actor network to propagate gradients effectively. This often results in unstable gradient updates or optimization failures, thereby limiting classical AC algorithms to generating either purely discrete or purely continuous action strategies. In the context of HVAC system control, the year-round optimization requires both minute-level fine-grained equipment control (e.g., fans and pumps) and longer-term mode selection (e.g., cooling, heating, ventilation). This hybrid task demands an algorithm that can handle both discrete and continuous action spaces, which poses significant challenges for existing algorithms. HDRL has the potential to address this limitation by decoupling action strategies with different characteristics and independently optimizing high-level and low-level

Table 3
Summary of state variables for the bottom and top agents.

Agent type	State variable [unit]	$x_{\min}$	$x_{\max}$
Bottom agent	Outdoor temperature [$^{\circ}\text{C}$]	-10	40
	Outdoor relative humidity [-]	0	100
	Operation time (0 or 1) [-]	0	1
	Occupancy [-]	0	110
	Hour [-]	0	23
	Internal heat gain [W]	0	8000
	Indoor temperature [$^{\circ}\text{C}$]	0	50
	Heat mass temperature [$^{\circ}\text{C}$]	0	50
	Indoor relative humidity [-]	0	100
	Indoor CO_2 concentration [ppm]	300	5000
	AHU inlet air temperature [$^{\circ}\text{C}$]	0	50
	AHU outlet air temperature [$^{\circ}\text{C}$]	0	50
	AHU inlet water temperature [$^{\circ}\text{C}$]	0	50
	AHU outlet water temperature [$^{\circ}\text{C}$]	0	50
	Air mass flow in AHU [kg/s]	0	2
	Water mass flow in AHU [kg/s]	0	1.5
	AHU power [kW]	0	5
	OAU power [kW]	0	4
	ASHP power [kW]	0	15
Top agent	Maximum outdoor temperature (current day) [$^{\circ}\text{C}$]	-10	40
	Minimum outdoor temperature (current day) [$^{\circ}\text{C}$]	-10	40
	Average outdoor temperature (current day) [$^{\circ}\text{C}$]	-10	40
	Month [-]	1	12
	Day [-]	1	31
	Day number from start of year [-]	1	365

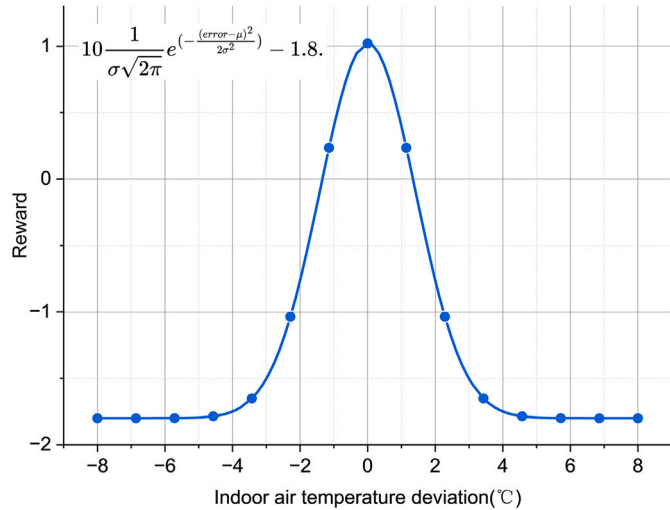


Fig. 8. Structure of indoor temperature control reward function.

policies using dedicated networks, making it well-suited for hybrid action spaces.

Building upon this potential, we propose an independently trained HDRL framework tailored to the hierarchical nature of HVAC system control. Low-level agents specialize in fine-tuned mode-specific control, thereby reducing exploration costs by focusing on local scenarios. High-level agents operate on fixed low-level policies, which optimize mode selection by capturing dynamic environmental changes. This approach simplifies training, reduces coupling complexity, and enables modular system deployment and maintenance, as policies can be updated independently.

Fig. 1 illustrates the proposed two-level HDRL framework. The top agent selects the daily operating mode based on outdoor conditions, thereby maintaining it throughout the day for optimal thermal comfort. The bottom agent, activated after mode selection, dynamically

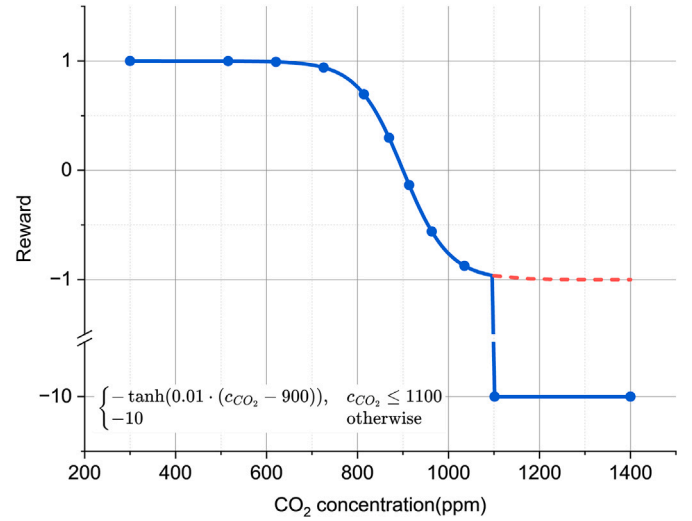


Fig. 9. Structure of indoor CO_2 concentration control reward function.

adjusts fan and pump frequencies every minute based on outdoor, indoor, and system parameters. This structure ensures indoor comfort and air quality while minimizing energy consumption, with the bottom agent handling continuous control and the top agent managing discrete control.

Fig. 2 illustrates the training process of the HDRL agent. First, the three bottom agents are trained using TD3 to optimize pump and fan frequencies. Once trained, the bottom agents become part of the environment for the top agent, which is then trained using PPO to select optimal modes. This hierarchical training results in agents that can ensure frequency regulation and mode selection.

3. Simulation platform development

A robust building energy system model is required as the RL environment to achieve the stable training of the agent and evaluate its control strategies. This model simulates indoor parameters and energy consumption responses to control changes. This section provides an overview of the model development.

3.1. Overview of the simulation platform

We developed a Python-based HVAC system simulation program with the following three key features: (1) The program can flexibly integrate various types of HVAC equipment and key components, thereby providing an evaluation platform that closely resembles real operating conditions. (2) The program captures the transient states of system components, which ensures that physical state changes correspond to control states, thereby facilitating the assessment of the impact of different control strategies on system operation. (3) The program features an open control interface, which allows different control logics (e.g., PID, MPC, DRL) to be imported as required to control system components. For a comprehensive derivation of the simulation platform, please see our previous work [10,41].

The computational flow is shown in Fig. 3. The program calculates control setpoints, flow rates, temperatures, and power consumption at adjustable time intervals. In this study, the time interval is configured to be 1 min.

3.2. Energy consumption model

The energy consumption includes four components: the Air Source Heat Pump (ASHP), pumps, fan of Outdoor Air Unit (OAU), and Air

Table 4
Summary of differences between the four methods.

Method	Fan and pump control (bottom level)	Operation mode selection (top level)	Descriptions
Method 1	PI control	Operation schedule based control	Baseline
Method 2	PI control	Outdoor temperature based control	PI control with mode selection
Method 3	DRL agent control	Operation schedule based control	DRL control
Method 4	DRL agent control	DRL agent control	the proposed HDRL control

Table 5
Hyperparameters in TD3 and PPO.

Hyperparameters	TD3	PPO
Learning rate	0.001	0.001
Discount factor	0.99	0.99
Batch size	100	64
Training steps	2,160,000	300,000
Clipping parameter	–	0.2
Soft update coefficient	0.05	–
Policy delay in TD3	2	–

Handling Unit (AHU). Energy consumed by the ASHP is calculated from the water temperature difference ΔT_w and flow rate Q_w (Eq. 3) as:

$$E_{\text{ASHP}} = \frac{C_w \rho_w Q_w \Delta T_w}{\text{COP}_{\text{ASHP}}}, \quad (3)$$

where, C_w represents the specific heat capacity of water at constant pressure, and COP_{ASHP} represents the Coefficient of Performance (COP) of the ASHP, which is assumed to be 3.15 based on design specifications caused by the lack of experimental data. Pump and fan power consumption and total system energy consumption are respectively calculated using:

$$E_{\text{fan,pump}} = \frac{Q \cdot \Delta P_s}{\eta}, \quad (4)$$

$$E_{\text{system}} = E_{\text{ASHP}} + E_{\text{pump}} + E_{\text{fan,oau}} + E_{\text{fan,ahu}}. \quad (5)$$

3.3. Indoor CO₂ concentration model for IAQ evaluation

Indoor CO₂ concentration is a common indicator to evaluate indoor air quality, modeled using the mass conservation equation. CO₂ concentration variation without air leakage is expressed as:

$$V_{\text{room}} \frac{dc_{\text{in}}}{dt} = Q (c_{\text{out}} - c_{\text{in}}) + G_{\text{CO}_2}, \quad (6)$$

where V_{room} represents the room volume (m³), Q represents the supply air flow rate, and c_{in} , c_{out} represent the CO₂ concentrations before and after a certain section, respectively (ppm). In this study, the outdoor CO₂ concentration is assumed to be constant at 350 ppm. G_{CO_2} represents CO₂ generation inside the room, which is calculated as:

$$G_{\text{CO}_2} = N_t \cdot q \cdot 10^6, \quad (7)$$

where N_t represents the number of occupants and q represents the CO₂ emission rate per person (assumed to be 30 L/h). When the ventilation system achieves a steady state, the differential equation is rewritten as a finite difference equation (Eq. 8):

$$c^t = \left[Q c^{t-1} + G_{\text{CO}_2}^{t-1} \right] \frac{\Delta T}{V_{\text{room}}} + c^{t-1}, \quad (8)$$

where t represents the current time step, $t-1$ represents the previous time step, and ΔT represents the time interval. This model describes the dynamic discrete CO₂ concentration, used to evaluate indoor air quality control effectiveness.

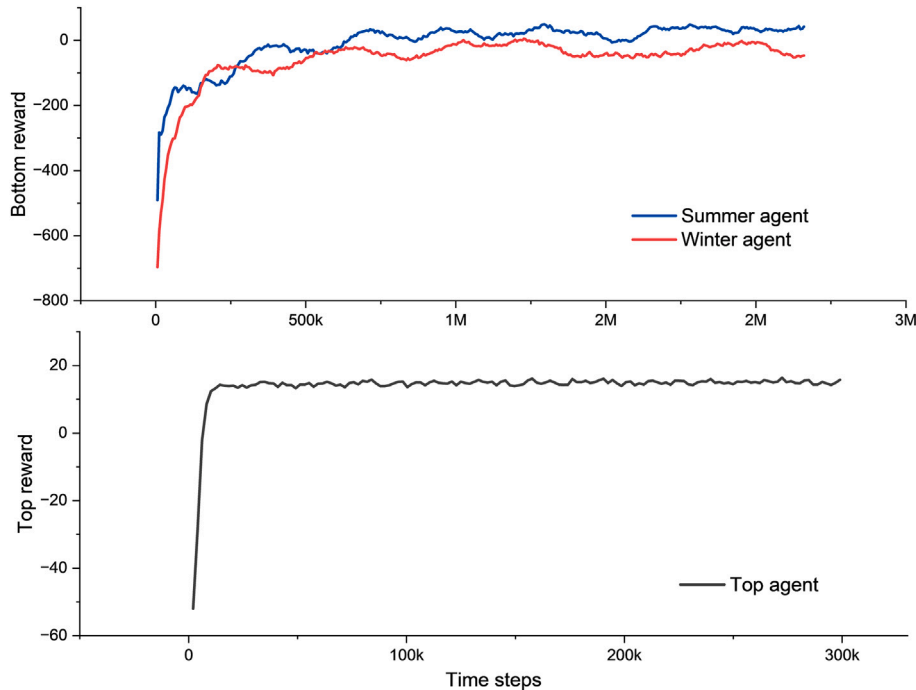
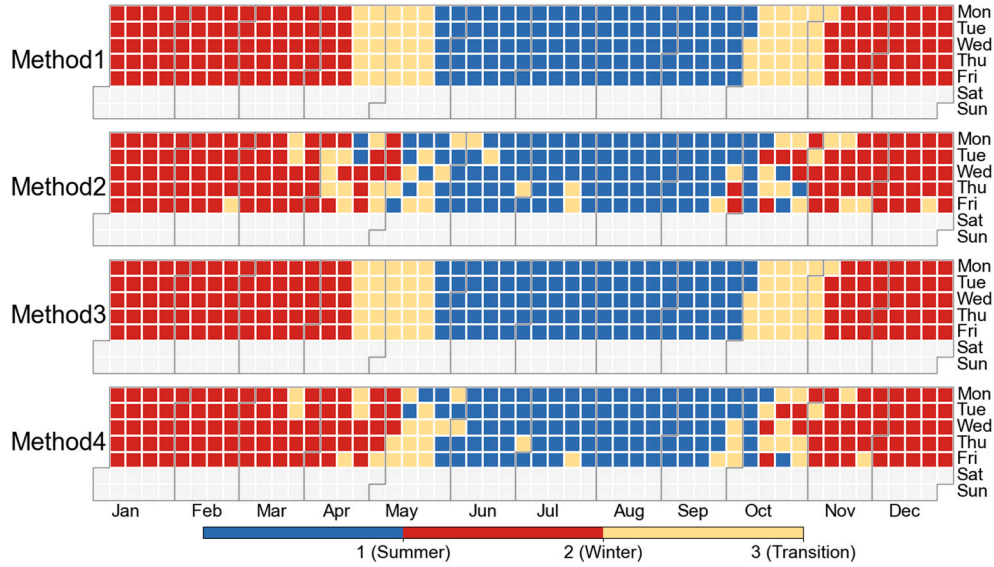


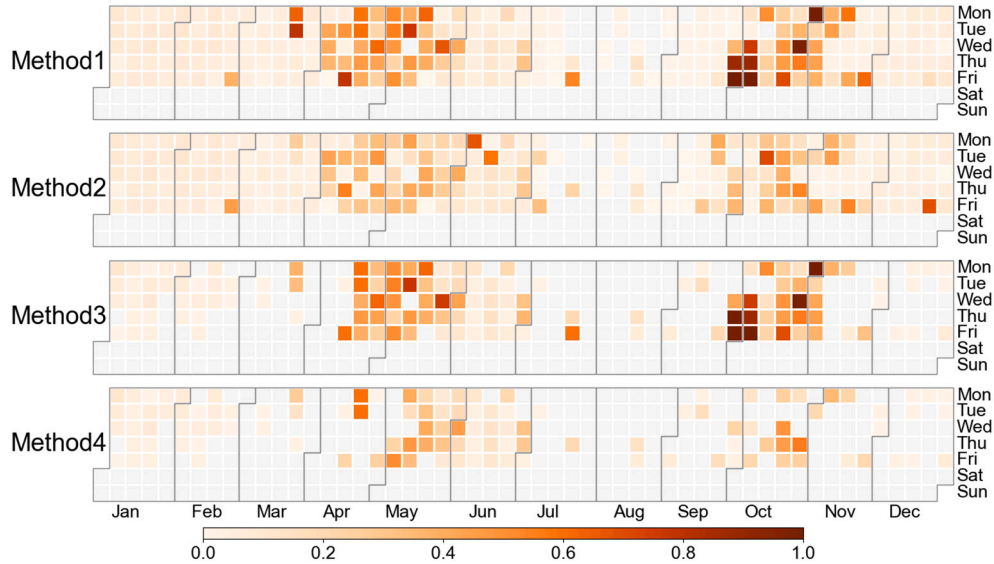
Fig. 10. Reward curve during the training process.

Table 6
Control achievement comparison for temperature, CO₂, and joint control for different methods.

Method	Temperature control		CO ₂ control		Joint control	
	Absolute time (min)	Time ratio (%)	Absolute time (min)	Time ratio (%)	Absolute time (min)	Time ratio (%)
Method 1	143,512	83.63	156,107	90.97	131,488	76.62
Method 2	148,513	86.55	153,274	89.32	133,025	77.52
Method 3	149,210	86.95	156,100	90.97	135,641	79.04
Method 4	155,530	90.64	155,700	90.73	141,750	82.60



(a) Daily operation mode selection chosen by Methods 1-4 throughout the year



(b) Daily temperature violation rates for Methods 1-4 throughout the year, with darker colors indicating higher violation rates.

Fig. 11. Calendar heatmap of operation mode selection and indoor temperature violation rate.

3.4. RC model for building energy demand and indoor air temperature calculations

The resistance-capacitance (RC) model is a widely used gray-box model to simulate thermal dynamics in buildings. Based on the electrical analogy to thermal physics, it provides fast load calculations while

capturing key physical dynamics. The EN ISO 13790 standard [42] comprehensively describes a simple hourly method using a five-resistance, one-capacitance (5R1C) network model. Jayathissa et al. [43] developed a Python-based 5R1C building simulator following this standard, which was suitable for single-zone buildings and demonstrated good performance in experiments.

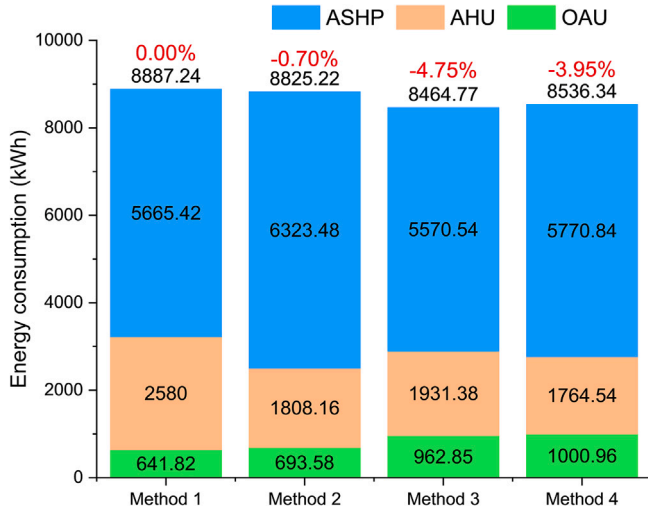


Fig. 12. System energy consumption.

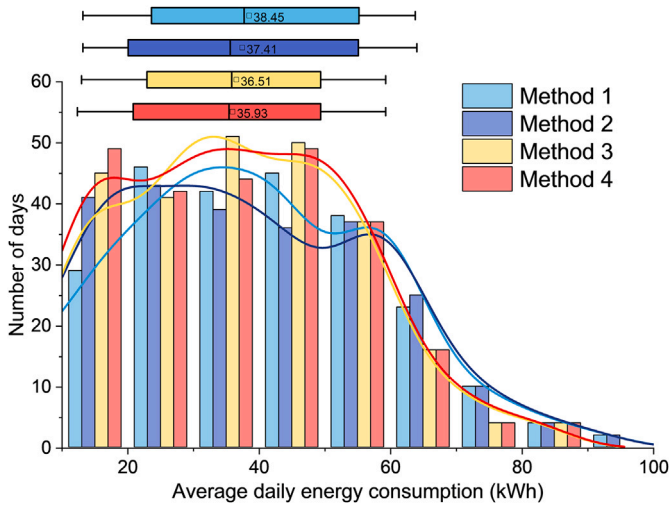


Fig. 13. Distribution of average daily energy consumption by control method.

We enhanced this model to a 5R2C version to improve accuracy for smaller time steps, adding an extra thermal capacitor (Fig. 4). The details of the calculation are provided in Appendix B. The validation of 5R2C model following the SHASE guidelines of the test procedure for the evaluation of building energy simulation tools is presented in Appendix C.

4. Case study

4.1. Distribution of the building and energy system

We select a lecture theater at a university in Tokyo, Japan, with a seating capacity of 110 people, as the case study for this research. Relevant building parameters are summarized in Table 2. An independent air conditioning system regulates the environment of the lecture theater, with the system layout shown in Fig. 5. The primary side comprises an Air Source Heat Pump (ASHP) water loop, while the secondary side includes a full-air system equipped with a total heat recovery unit.

4.2. Test conditions

We used the typical meteorological year data from Chiyoda, Tokyo (2007–2012) (Fig. 6) as the input for outdoor conditions during training

and validation. The occupancy schedule was based on the EnergyPlus large lecture theater schedule (Fig. 7). Maximum lighting and equipment power was set at 10 W/m², with a heat gain of 60 W per occupant. The CO₂ emission rate per person was 30 L/h. Using this schedule, we calculated the indoor heat load and total CO₂ production.

4.3. Design of state space

The observation space of the agent could theoretically be very large, allowing comprehensive understanding of the environment. However, a larger state space complicates learning, and it is difficult to fully capture operational data for all HVAC components. Considering these factors, the state observations for the bottom and top agents are summarized in Table 3. Furthermore, data acquisition feasibility is considered. The bottom agent has 19 observation items, whereas the top agent has six observation items, thereby covering indoor parameters, outdoor weather, HVAC operational data, and time information.

It is important to note that different state observations vary in units and scales. We applied Min-Max normalization to scale all values between 0 and 1 to ensure proper weighting during training and improve efficiency and stability:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (9)$$

where $x_{\min}$ and $x_{\max}$ represent the lower and upper bounds of each state variable.

4.4. Design of the action space

The bottom agent controls the inverter frequencies of the system's fans and pumps. The action space is three-dimensional, and it consists of continuous values for the frequency ratios of the AHU fan, OAU fan, and pump (ranging from 0 to 1).

The top agent selects the optimal daily operating mode, with three discrete values: Mode 1 (summer), Mode 2 (winter), and Mode 3 (transition). Then, the bottom agent executes specific inverter controls for the selected mode. The hybrid nature of the action space makes training challenging for traditional DRL algorithms: this is the primary reason to adopt a hierarchical approach for training the agents.

4.5. Design of the reward function

The reward function defines the objectives of DRL. We designed three sub-reward functions, which are combined using weighted parameters to enable the bottom agent to consider indoor temperature, air quality, and energy consumption. The reward for the top agent is based on the average indoor temperature of each day, which guides the mode selection.

4.5.1. Reward based on indoor temperature control

We assume that the HVAC system maintains indoor temperatures of 22 °C in winter and 26 °C in summer, with the temperature deviation error calculated for each time step. The comfort range for temperature deviation is ± 2 °C. A Gaussian distribution (Eq. 10) with a mean of 0 (indicating zero target temperature deviation) and a standard deviation of $\sqrt{2}$ is used for defining the temperature control reward (Eq. 11).

As shown in Fig. 8, the reward ranges from -1.8 to 1 . The reward increases as the temperature approaches the target, which encourages exploration during optimization. Furthermore, restricting the value range of the reward function facilitates gradient descent calculation during network updates. The feasibility of this type of reward function has been validated in previous studies [21,44,45].

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \quad (10)$$

$$R_{temp} = 10 \frac{1}{\sqrt{2}\sqrt{2\pi}} \exp\left(-\frac{(error-0)^2}{2 \cdot 2}\right) - 1.8. \quad (11)$$

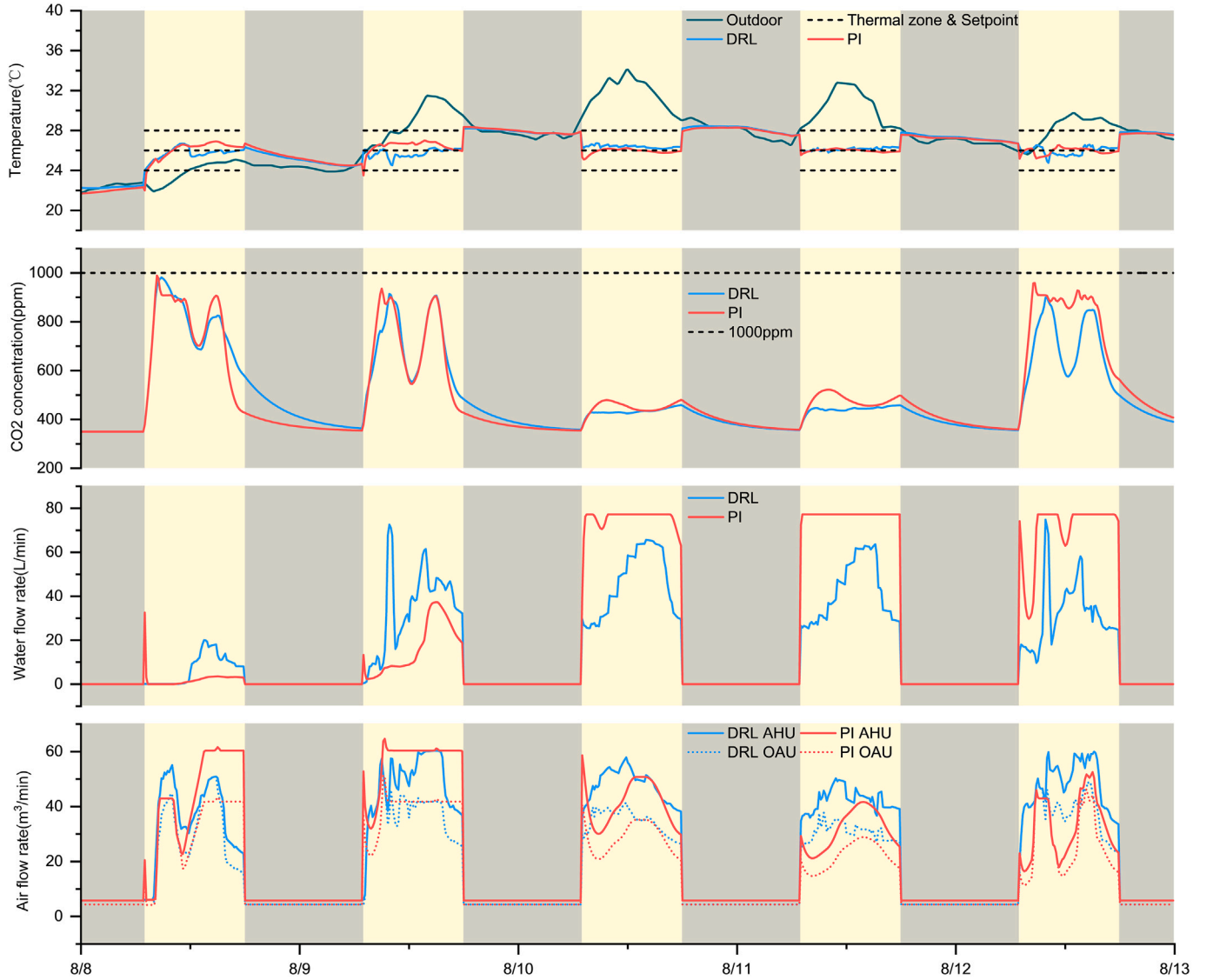


Fig. 14. Changes in indoor temperature and CO₂ concentration and system operation in typical cooling scenario.

4.5.2. Reward based on CO₂ concentration control

We set the target CO₂ concentration at 900 ppm (slightly below 1000 ppm) to ensure good air quality. A positive reward is given when CO₂ is below 900 ppm, with the reward increasing as CO₂ decreases; however, it gradually levels off to reflect a trade-off between fresh air and increased fan energy consumption. When CO₂ exceeds 900 ppm, a penalty is applied, which is capped to maintain stable training. The reward function is designed using a hyperbolic tangent function (Eq. 12, Fig. 9) to meet these requirements because it allows smooth reward scaling and is effective for stable learning.

$$R_{CO_2} = \begin{cases} -\tanh(0.01 \cdot (c_{CO_2} - 900)), & \text{if } c_{CO_2} \leq 1100 \\ -10 & \text{otherwise} \end{cases} \quad (12)$$

4.5.3. Reward for reducing energy consumption

The energy reward is defined as the negative value of total consumption to minimize energy consumption while maintaining comfort (Eq. 13). Based on the simulation results of traditional control methods,

Min-Max normalization is applied to ensure R_{energy} has the same magnitude as R_{temp} and R_{CO_2} , which aids in the balanced exploration.

$$R_{energy} = -(\hat{E}_{ASHPs} + \hat{E}_{AHU} + \hat{E}_{OAU}). \quad (13)$$

Three weights w_1 , w_2 , and w_3 adjust the contribution of comfort, air quality, and energy consumption to the bottom reward (Eq. 14). The weights w_1 , w_2 , and w_3 sum to 1, where a larger weight reflects a greater emphasis on that specific objective. R_{temp} , R_{CO_2} , and R_{energy} are normalized to the consistent range and magnitude, and therefore, these weights can be adjusted between 0 and 1, thereby ensuring flexibility and robustness in reward function design.

$$R_{bottom} = w_1 R_{temp} + w_2 R_{CO_2} + w_3 R_{energy}. \quad (14)$$

4.5.4. Reward for operation mode selection

The top agent selects the optimal HVAC mode for each day using Eq. (15). Since the objective of the top agent is simpler with a low-dimensional discrete action space, a straightforward reward function is used to reduce training costs without sacrificing quality.

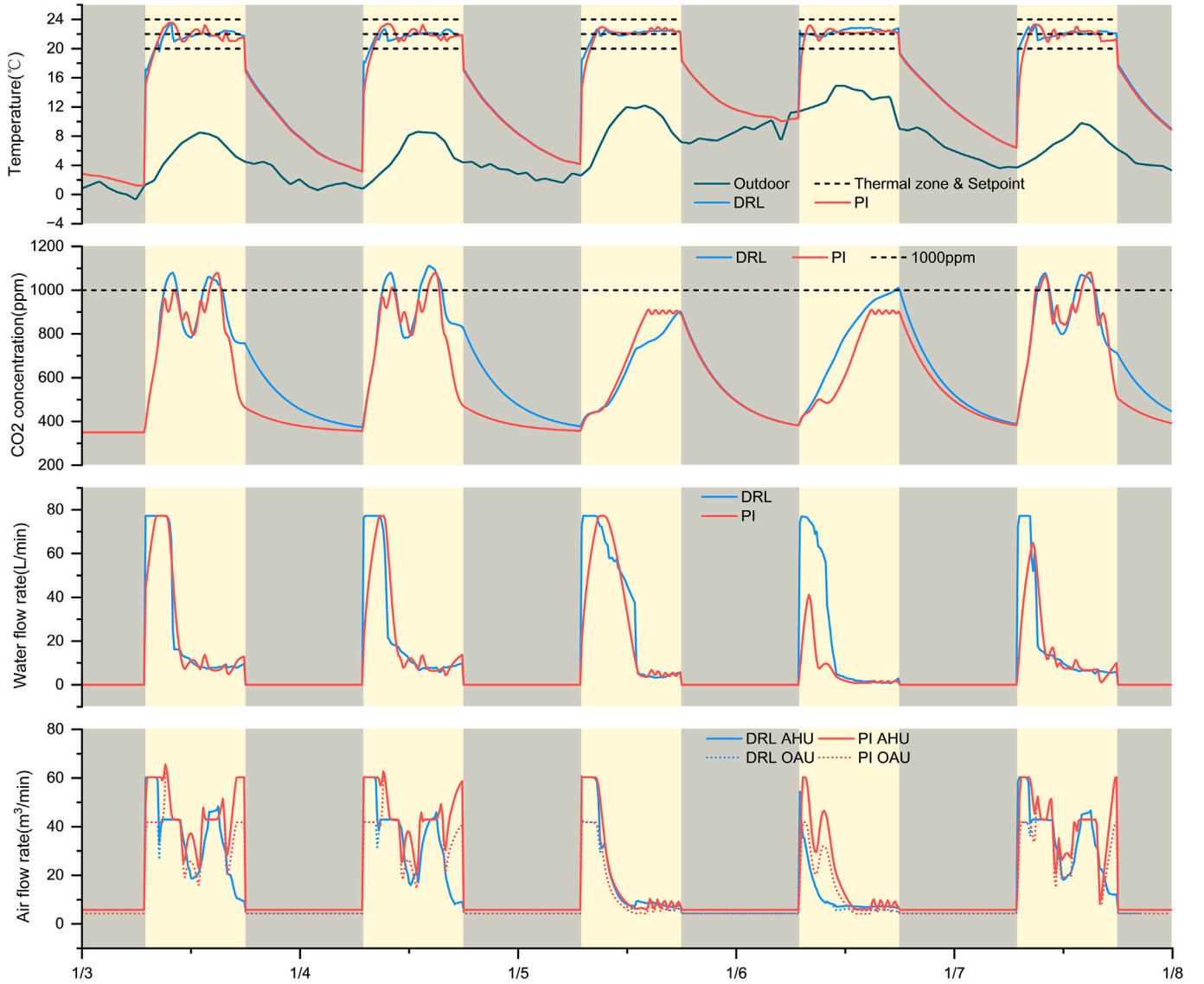


Fig. 15. Changes in indoor temperature and CO₂ concentration and system operation in typical heating scenario.

The reward is determined based on the indoor temperature performance for the selected mode. In heating and cooling modes, rewards increase as the temperature gets closer to the ideal value. In the transition mode, maintaining the temperature between 20 and 26 °C earns the full reward, encouraging energy-saving operation under mild conditions. A significant penalty is applied for inappropriate mode selection to guide the agent effectively.

$$R_{\text{top}} = \frac{1}{T_{\text{operation}}} \sum_{t=1}^{T_{\text{operation}}} \begin{cases} 1 - 0.6 \cdot |t_{\text{room}} - 26|, & \text{if Mode = 1 and } 24 \leq t_{\text{room}} \leq 28, \\ -5, & \text{otherwise (Mode = 1)} \\ 1 - 0.6 \cdot |t_{\text{room}} - 22|, & \text{if Mode = 2 and } 20 \leq t_{\text{room}} \leq 24, \\ -5, & \text{otherwise (Mode = 2)} \\ 1, & \text{if Mode = 3 and } 20 \leq t_{\text{room}} \leq 26, \\ -2, & \text{otherwise (Mode = 3)} \end{cases} \quad (15)$$

4.6. Numerical experiment setting

All simulations in this study were conducted on a per-minute basis, using one year of training and testing data. To evaluate the

effectiveness of the HDRL method, four HVAC control strategies were compared:

- **Method 1 (PI control method):** The rule-based PI controller is used as a baseline and operates based on design specifications. The PI controller adjusts fan and pump frequencies at each time step. The OAU fan is controlled to keep indoor CO₂ levels below 900 ppm, whereas the AHU fan frequency is adjusted to maintain indoor temperatures at 26 °C in summer and 22 °C in winter. The pump frequency is adjusted based on the supply air temperature. During the transition season, full fresh air operation is used by closing the AHU water valve to cool the room using outdoor air. PI gains are tuned through simulation for optimal control. The system does not automatically select the optimal mode; instead, manual adjustments follow the operation schedule from Japanese energy-saving guidelines.
- **Method 2 (PI control with mode selection):** The method built on Method 1, still using PI controllers to adjust fan and pump frequencies, with identical control targets. Unlike Method 1, Method 2 introduces an automatic mode selection scheme based on the daily maximum outdoor temperature. The cooling mode is activated if the

temperature is above or equal to 24 °C, the heating mode is activated when the temperature is below 22 °C, and the transition mode is activated for temperatures in between.

- **Method 3 (DRL control):** In this method, three trained agents replace the PI controllers from Method 1 to control the fan and pump frequencies for each mode. The agents provide control signals based on observations of the HVAC system, indoor, and outdoor environments. To align with Method 4, these agents are referred to as bottom agents: a summer agent, winter agent, and transition agent. Similar to Method 1, the daily operating mode is determined using a fixed schedule.
- **Method 4 (HDRL control):** The complete control approach proposed in this study uses two hierarchical agents trained with the HDRL framework. The bottom agents, similar to those in Method 3, adjust fan and pump frequencies. In addition, a top agent selects the optimal operating mode for each day. HDRL control automatically determines the appropriate mode based on daily outdoor meteorological data, which makes it an upgraded version of Method 3.

Table 4 summarizes the differences between the four control methods.

4.7. Training processes

We used the OpenAI Gym framework [46] to create a custom RL environment, which encapsulates the HVAC control system and the 5R2C building thermal model (detailed in Section 3). Agent training was implemented using the TD3 and PPO algorithms from the open-source library Stable Baselines3 [47] on PyTorch and accelerated with an RTX 3090 GPU. Table 5 summarizes the hyperparameter settings.

During training, we monitored the performance improvement of the agent and cumulative reward convergence. Fig. 10 shows training curves for both bottom and top agents. The bottom agents were trained for 1500 episodes of 1440 time steps each (minutes), with random initial times to improve generalization. The top agent was trained for 10,000 episodes of 30 time steps each (days). The cumulative reward of the top agent stabilized after 30,000 time steps, whereas the bottom agents peaked around 750,000 time steps, with fluctuations near convergence thereafter. Overall, the top agent converged faster because it made discrete daily decisions, while the bottom agents optimized complex minute-level actions. The training curves align with design expectations and typical DRL training behavior.

5. Results and discussion

We compared the four control methods introduced in Section 4.6 over a one-year testing period. The analysis focused on indoor temperature control, CO₂ concentration, and energy consumption across the annual operating cycle to evaluate the performance of each method.

5.1. Indoor temperature and CO₂ concentration control

For indoor temperature control, comfort ranges are defined as 24–28 °C in summer, 20–24 °C in winter, and 20–26 °C during transitional seasons, with the time spent within these ranges termed temperature compliance time. For CO₂ control, a 1000 ppm threshold defines acceptable air quality, with air quality compliance time referring to periods below this level. Joint control time denotes the duration meeting both temperature and air quality criteria simultaneously. Table 6 lists control time differences for indoor temperature and CO₂ concentration over the entire year of operation for the four control methods.

For temperature control, Method 1 (PI control) performs the worst, which achieves a compliance time of 143,512 min and a compliance rate of 83.63 %. Method 2 shows slight improvement, which indicates that rule-based mode selection better aligns with seasonal changes. Method 3 further improves with a compliance rate of 86.95 %, suggesting that the trained bottom agents effectively interpret the temperature reward

and make better decisions based on environmental observations. Method 4 achieves the best temperature control, with 155,530 min of compliance and a rate of 90.64 %, significantly outperforming the other methods. This demonstrates that the top agent of Method 4 further enhances temperature control by selecting optimal modes based on outdoor conditions, building on the improved performance of the bottom agents.

Fig. 11 presents a calendar heatmap of mode selection and temperature violation rates over the year. Methods 1 and 3 follow a fixed mode selection schedule, and although Method 3 performs better during cooling and heating periods (lighter colors), both methods show numerous temperature violations during the transition seasons, thereby emphasizing the importance of mode selection. Method 2, which uses preset rules for mode selection, performs better in transition seasons than that in Methods 1 and 3; however, it struggles with mismatches between mode and actual weather conditions. This results in more severe violations on certain days in February and December compared to that of Methods 1 and 3. Method 4 uses the top agent to intelligently select modes based on state changes, achieving better performance, particularly during the transition seasons. The adaptive mode selection of the top agent, coupled with the effective control of the bottom agent, makes Method 4 the best performer in temperature control across the year.

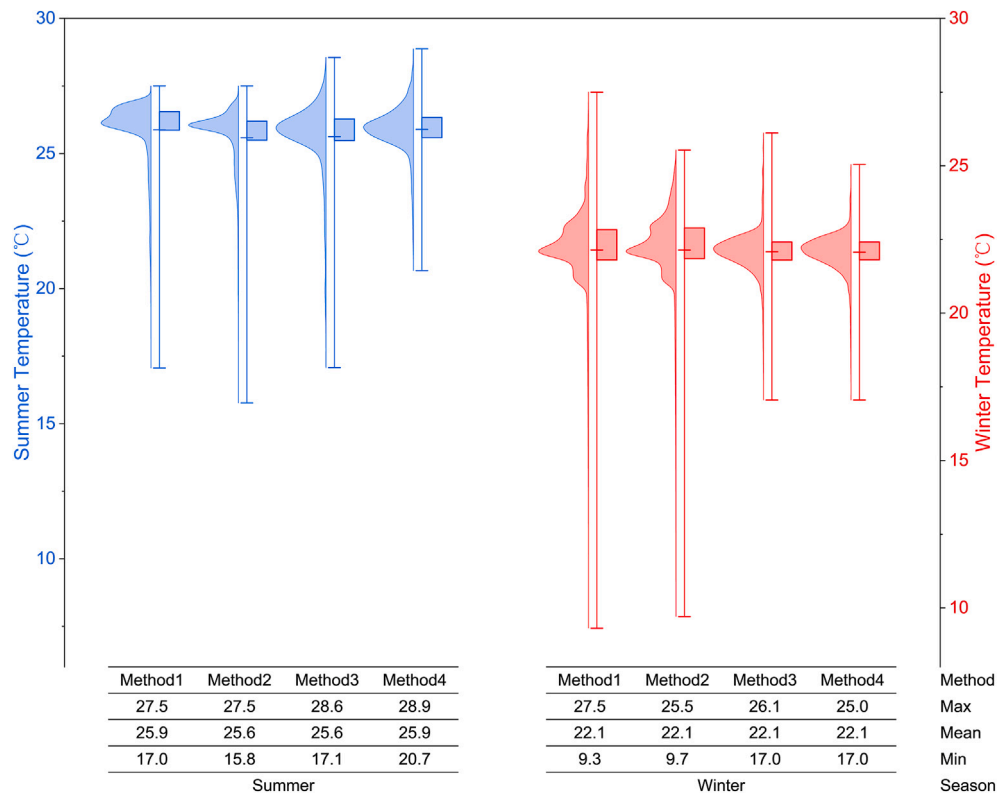
The CO₂ control results in Table 6 reveal that, Methods 1 and 3 achieve the same CO₂ compliance rate of 90.97 %, which is the highest among all methods. Methods 2 and 4 have slightly lower compliance rates at 89.32 % and 90.73 %, respectively. This slight decrease can be attributed to the flexible mode selection in Methods 2 and 4, which assign cooling or heating modes instead of the more CO₂-efficient transition mode on certain days, thereby prioritizing temperature control over CO₂ levels. However, these differences are minor, and overall, the CO₂ control performance across the four methods is considerably similar.

For combined temperature and CO₂ control, Table 6 shows an increasing trend in joint compliance time for Methods 1 to 4. Method 1 has a joint compliance rate of 76.62 %, with Method 2 slightly improving it to 77.52 %. Introducing RL agents in Methods 3 and 4 significantly enhances joint control performance, with Method 3 reaching 79.04 % and Method 4 further increasing to 82.60 %. These results indicate that well-trained agents provide better control signals, enabling a more comprehensive balance of temperature and CO₂ control. These results validate the goal of balancing both factors through reward function design, and highlight the superior performance of Method 4 as a hierarchical agent control approach for complex tasks.

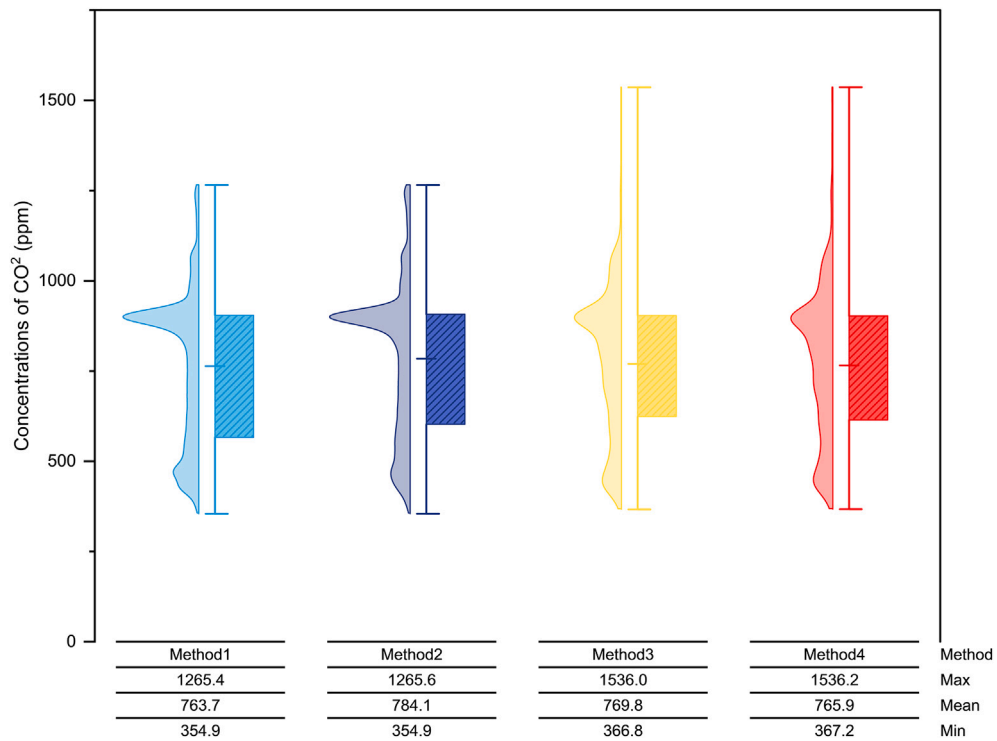
5.2. Energy consumption

Fig. 12 shows the annual energy consumption of the HVAC system across different methods. Method 1, which serves as the baseline, has the highest total energy consumption at 8887.24 kWh. Method 2 slightly reduces energy consumption by employing flexible mode selection. Method 3, which uses a fixed schedule for mode selection, significantly reduces energy consumption by about 4.75 %, demonstrating that well-trained agents can generate more efficient fan and pump operation strategies. Method 4 consumes 8536.34 kWh, which is slightly higher than Method 3 but still saves approximately 3.95 % compared to Method 1. The increased consumption of Method 4 is attributed to its flexible mode selection, which increased the number of cooling and heating days. This led to improved control of both temperature and CO₂ (as summarized in Section 5.1) at the cost of slightly increased energy use.

Fig. 13 highlights the energy consumption patterns across different methods from the perspective of average daily consumption. Method 1 shows high average daily energy use and frequent high-consumption days. Although Method 2 improves comfort through rule-based mode selection, it exhibits a similar frequency of high-consumption days as Method 1, reflecting the limited energy-saving potential of PI control. In contrast, Methods 3 and 4, which utilize DRL agents, achieve lower



(a) Indoor air temperature distribution during summer and winter operation.

(b) Indoor CO₂ concentration distribution during year-round operation.Fig. 16. Distribution of indoor air temperature and CO₂ concentration during the operation period.

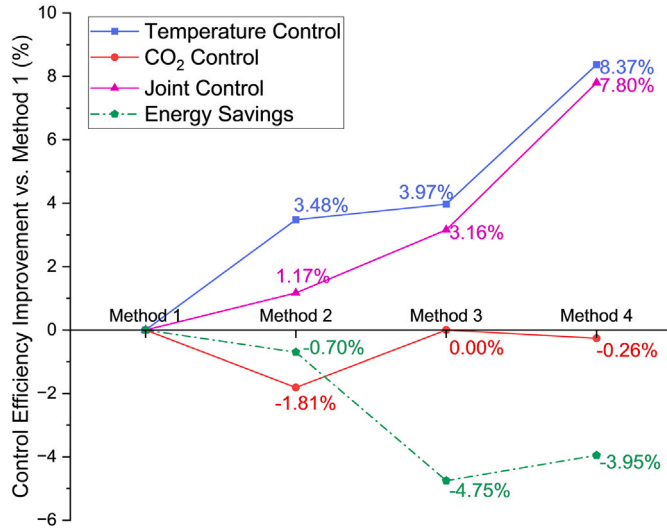


Fig. 17. Control efficiency improvement and energy savings compared to Method 1.

average daily consumption and a more concentrated distribution, indicating superior multi-objective performance by effectively balancing comfort and energy efficiency. The total annual cooling and heating days are 211 (Method 1), 221 (Method 2), 211 (Method 3), and 224 (Method 4). While Method 4 shows slightly higher cumulative consumption than Method 3, it compensates with 13 additional active days and improved comfort compliance, making it the most balanced and effective control strategy.

5.3. Detail visualization analysis

We selected typical cooling and heating scenarios to illustrate the variations in control strategies during actual operations, demonstrating the control effectiveness of the proposed control method.

Figs. 14 and 15 show detailed variations in indoor temperature, CO₂ concentration, and system operation under cooling and heating scenarios for both the bottom DRL control and the PI control method. The dashed lines indicate the comfortable temperature range: 24–28 °C in summer and 20–24 °C in winter, and the CO₂ concentration limit of 1000 ppm during work hours. The results confirm that, while PI method maintains relatively stable temperature control, the DRL agents demonstrate more adaptive actions in response to complex outdoor temperature changes and indoor occupancy, with a stronger focus on energy efficiency.

Under typical PI control, control targets of the components are fixed. From August 10 to 12, the pump operates at a high frequency to maintain a supply air temperature of 16 °C, which leads to high energy consumption. Since the indoor temperature stays around 26 °C, the AHU fan frequency remains stable, thereby leading to low airflow. In contrast, DRL agents adjust pump frequency based on outdoor temperature and occupancy, which increases airflow to maintain both temperature and CO₂ levels. On August 9, PI control maintains the pump frequency low in the morning because of low outdoor temperatures. When external load and occupancy increase, the pump frequency fails to adjust in time, thereby causing overheating. Although the fan is nearly fully open, the pump remains underutilized because the indoor temperature has not exceeded 27 °C, the condition required for changing the pump setpoint, thereby causing insufficient cooling. However, DRL agents adapt airflow and pump frequency promptly to meet rising loads, preventing overheating. During this summer week, PI control consumes 255.4 kWh, while

DRL control consumes 252.7 kWh, thereby saving 1.5 % energy with better performance.

Similar conclusions can be drawn from winter results. During winter, the DRL agents optimize control actions based on comprehensive rewards, minimizing fresh air intake to reduce external load while maintaining CO₂ below 1000 ppm. At the initial load stage, agents start the pump earlier to heat the room quickly, and during periods with high internal heat generation, they shut down the pump promptly (e.g., January 3, 4, and 7) to prevent overheating and energy waste. The agents learn the work schedule and room thermal inertia, which is difficult for PI control to achieve. During this winter week, PI control consumes 265.0 kWh, whereas DRL control uses 238.2 kWh, thereby achieving an energy saving of about 10.1 %.

Overall, the DRL agents make decisions by comprehensively observing the environment and balancing reward objectives. Well-trained agents determine optimal fan and pump frequency combinations for achieving better energy efficiency while maintaining comfort and air quality. This highlights the significant advantages of DRL agents in flexibility and adaptability to changing conditions.

5.4. Summary of control performance

Method 1 represents typical PI control for regulating pump and fan frequencies in the HVAC system. Method 2 enhances Method 1 with flexible daily mode selection. Methods 3 and 4 introduce RL: Method 3 replaces PI controllers with DRL agents, which follow a fixed schedule for mode selection, whereas Method 4 uses a two-layer DRL agent for flexible mode selection. This subsection summarizes the performance of these four methods in temperature control, CO₂ control, and energy consumption.

- Indoor temperature control: Fig. 16 shows that all four methods maintain the indoor temperature within the comfortable range for most of the working hours throughout the year (24–28 °C in summer and 20–24 °C in winter). Methods 3 and 4 exhibit smaller temperature deviations, because of the precise control by the DRL agents. Method 4, in particular, has the smallest deviation, which indicates that the top agent selects appropriate modes, thereby avoiding temperature mismatches. The maximum temperatures in Methods 3 and 4 occasionally exceed 28 °C because agents reduce HVAC output during unoccupied midday periods with low CO₂ and high outdoor loads to save energy, thereby leading to slight overheating. However, such cases are rare throughout the year.
- In terms of CO₂ control (Fig. 16), Methods 3 and 4 show higher maximum indoor CO₂ concentrations due to concessions made by the agents to prioritize temperature regulation or energy savings. However, the average CO₂ concentration across all four methods is similar, which indicates that air quality requirements are still met. Overall, Method 4 achieves a better balance between temperature and CO₂ control.
- Fig. 17 compares the control performance and energy savings of each method, using Method 1 as the baseline. Method 2, with logical mode selection, and Method 3, using agents to regulate pump and fan frequencies, both show moderate control improvements. Method 4, with hierarchical agents, achieves the highest combined control improvement at 7.8 %. Although Method 3 achieves the best energy savings, the extended cooling and heating periods of Method 4 slightly reduce the energy-saving effect. However, Method 4 provides a more balanced and satisfactory performance in both temperature and CO₂ control.

6. Conclusions, limitations and future work

We proposed an HDRL agent-based control method for HVAC systems that comprehensively considers indoor temperature, CO₂ concentration,

and energy consumption. The approach employs bottom agents for optimizing fan and pump control, while the top agent optimizes mode selection. A Python-based HVAC system model and a 5R2C thermal model were developed for performing minute-level simulation control, thereby forming the basis for the hierarchical reinforcement learning-based optimization method. We trained both single-layer and hierarchical agents and evaluated the proposed scheme by designing four control methods, which compare their performance over a full year of operation. The main conclusions are as follows:

- To ensure reliable and stable agent training, we designed a Gaussian distribution-based reward for temperature control, hyperbolic tangent-based reward for CO₂ control, and normalized energy consumption reward for the bottom agents to ensure reliable and stable agent training. For the top agent, a temperature-based mode selection reward was designed. Simulation results indicated that these reward functions effectively facilitated HVAC multi-objective optimization. The three adjustable weight parameters enabled users to customize the balance between different control objectives, thereby enhancing both the flexibility and applicability of the reward functions.
- The TD3 and PPO algorithms were used for training agents at different levels, resulting in an HDRL-based HVAC optimization method. The two layers of agents managed the variable frequency control of fans and pumps and mode selection, respectively. This hierarchical approach overcame the challenges faced by a single agent in learning within a hybrid action space, thereby making the application of DRL algorithms more feasible for practical HVAC control.
- We designed four different control methods and verified the effectiveness of the HDRL method through comparative analysis. The annual simulation results indicate that the HDRL method responds more flexibly to changes in indoor occupancy and outdoor weather, effectively maintaining indoor temperature within the comfort range, controlling air quality, and minimizing energy consumption. Compared to classical PI control, the HDRL method increased the temperature compliance rate by 8.37 % and the temperature-CO₂ joint compliance rate by 7.80 %, while saving 3.95 % of energy, with a CO₂ compliance rate maintained at the baseline level.

Despite the advantages of the proposed control method, there are several limitations that should be addressed in future research:

- Training of the DRL agents and validating their deployment depend on environmental observations. In this study, we assumed that meteorological data, HVAC system operation data, and indoor environmental data could be fully obtained, enabling the agents to learn the latent patterns of environmental changes and make optimal decisions. This highlights the importance of Internet of Things (IoT) and intelligent sensors.
- The design of reward functions is critical for control performance but remains a complex and time-consuming task. This study did not fully explore reward tuning because of time constraints. Generative Adversarial Imitation Learning (GAIL) presents a promising alternative by automatically deriving reward signals from expert demonstrations. Instead of relying on explicitly defined rewards, GAIL learns policies directly from state-action distributions, leading to faster convergence, higher data efficiency, and improved multi-objective optimization, as recommended by Liu et al. [48].
- Although HDRL effectively addresses the hybrid action space problem in HVAC control, the training process of the framework requires sequential training of bottom and top agents, which results in high computational overhead. Future research can explore multi-agent deep reinforcement learning (MADRL) as an ideal solution to accelerate convergence through parallel computation [49]. However, it is important to note that MADRL still faces significant challenges when

dealing with hybrid action spaces, requiring further investigation to fully address these issues.

- Because of a lack of measured data, the year-round optimization results of this study are simulation based. Although comprehensive simulation models were developed to ensure that the agents were trained and validated in environments resembling real-world systems, strategies derived from simulations often underperform in practical deployments, as highlighted by [50,51]. Enhancing the robustness of sim-to-real transfer remains a critical challenge. Potential solutions include integrating domain-adaptive transfer learning [52] into DRL and retraining the agent after deployment to better adapt to real-world conditions [53].

CRediT authorship contribution statement

Chenxin Liao: Writing – original draft, visualization, software, and methodology. **Shohei Miyata:** Writing – review & editing, supervision, and funding acquisition. **Ming Qu:** Writing – review & editing and supervision. **Yasunori Akashi:** Writing – review & editing, supervision, and funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Detailed methodologies of the AC methods

The AC method involves simultaneous updates to the Critic and Actor networks using the TD error. The Critic and Actor networks are represented by:

$$Q^\pi(s, a; \phi) = \mathbb{E}[R_t | s_t = s, a_t = a; \phi], \quad (\text{A.1})$$

$$\pi(a|s; \theta) = p(a_t = a | s_t = s; \theta). \quad (\text{A.2})$$

where ϕ and θ represent the parameters of the Critic and Actor networks, respectively. The Critic minimizes the Temporal Difference (TD) error δ :

$$\delta = r_t + \gamma Q(s_{t+1}, a_{t+1}; \phi) - Q(s_t, a_t; \phi), \quad (\text{A.3})$$

and its loss function is defined as:

$$\mathcal{L}(\phi) = \mathbb{E}[\delta^2]. \quad (\text{A.4})$$

The Actor aims to maximize the expected cumulative reward, represented by:

$$\mathcal{J}(\theta) = \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta} [Q^\pi(s, a; \phi)], \quad (\text{A.5})$$

The Critic and Actor are updated via gradient descent and ascent, respectively:

$$\phi \leftarrow \phi - \alpha \nabla_\phi \mathcal{L}(\phi), \quad \theta \leftarrow \theta + \beta \nabla_\theta \mathcal{J}(\theta), \quad (\text{A.6})$$

where α and β represent the learning rates for the Critic and Actor networks, respectively.

A.1. TD3 algorithm details

The TD3 algorithm maintains two Critic networks $Q(s, a; \phi_1)$ and $Q(s, a; \phi_2)$, as well as an Actor network $\mu(s; \theta)$. The Critic networks estimate the action-value function, while the Actor determines the optimal action for a given state. The objective of the Critic networks is to minimize the Bellman error, which is defined as follows:

$$\mathcal{L}(\phi_i) = \mathbb{E}_{(s, a, r, s')} \left[(Q(s, a; \phi_i) - y)^2 \right], \quad \text{for } i = 1, 2, \quad (\text{A.7})$$

where the target value y is computed as:

$$y = r + \gamma \min_{i=1,2} Q(s', a'; \phi'_i). \quad (\text{A.8})$$

Here, $a' = \mu(s'; \theta') + \epsilon$, and $\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$. ϕ'_i and θ' represent the parameters of the target Critic and Actor networks, respectively, and γ represent the discount factor. The addition of noise ϵ to the target action introduces target policy smoothing, which helps reduce overestimation bias.

The Actor network is updated by maximizing the expected value as estimated by one of the Critics:

$$\mathcal{J}(\theta) = \mathbb{E}_s [Q(s, \mu(s; \theta); \phi_1)]. \quad (\text{A.9})$$

To improve stability, TD3 delays the update of the Actor network. Specifically, the Actor is updated less frequently than the Critic, typically every two or more iterations. In addition, target networks are updated using a soft update mechanism:

$$\phi'_i \leftarrow \tau \phi + (1 - \tau) \phi'_i, \quad \theta' \leftarrow \tau \theta + (1 - \tau) \theta', \quad (\text{A.10})$$

where τ represents a small positive constant (e.g., $\tau = 0.005$) that controls the rate of updates.

A.2. PPO algorithm details

The core idea of PPO is optimizing a surrogate objective that restricts policy updates to a certain range, which can be achieved in two ways: PPO-Clip and PPO-Penalty. In PPO-Clip, the objective function uses a clipping mechanism:

$$\mathcal{L}_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (\text{A.11})$$

where $r_t(\theta) = \frac{\pi(a_t|s_t; \theta)}{\pi(a_t|s_t; \theta_{\text{old}})}$ represents the probability ratio between the new and old policies, $\hat{A}_t$ represents the advantage function, and ϵ is a hyperparameter that controls the clipping range (typically $\epsilon \in [0.1, 0.3]$). The clipping function ensures that the probability ratio $r_t(\theta)$ remains within the range $[1 - \epsilon, 1 + \epsilon]$, which limits the magnitude of policy updates and ensuring learning stability.

In PPO-Penalty, a penalty term is added to the objective function to explicitly constrain the KL divergence between the new and old policies. In this study, we use the pre-defined PPO algorithm provided by the Stable Baselines 3 library to train the agent, where the PPO-Clip variant is adopted. Owing to space limitations, the other variant of PPO, namely PPO-Penalty, is not discussed in detail.

Appendix B. Calculation procedure of the 5R2C model

The 5R2C model can be formulated using the node potential method for deriving the node temperature equations, similar to Kirchhoff's laws in electrical circuits. Assuming the sum of heat passing through each independent node is zero, the equations can be expressed in the matrix form as:

$$H \cdot T = I, \quad (\text{B.1})$$

where H represents the thermal conductance matrix, T represents the node temperature vectors, and I represents the heat current source

vectors. For the 5R2C network, Eq. (B.1) can be converted into the specific form:

$$\begin{bmatrix} H_{em} + H_{ms} & -H_{ms} & 0 \\ -H_{ms} & H_{ms} + H_w + H_{is} & -H_{is} \\ 0 & -H_{is} & H_{is} + H_{ve} \end{bmatrix} \begin{bmatrix} T_m \\ T_s \\ T_{air} \end{bmatrix} = \begin{bmatrix} \Phi_m - \Phi_c + H_{em} T_e \\ \Phi_{st} + H_w T_e \\ \Phi_{ia} + \Phi_{HC} + H_{ve} T_{sup} - \Phi_{air} \end{bmatrix}, \quad (\text{B.2})$$

where T_m , T_s , T_{air} , T_e , and T_{sup} represent the temperature of the thermal mass, temperature of the internal surfaces of the room, temperature of the room air, external air temperature, and temperature of the ventilation air, which is assumed to be equal to the external air temperature, respectively. Further, H_{ve} , H_{is} , H_w , H_{em} , and H_{ms} represent the ventilation conductance which can be set to 0.0001 W/K by default when the room is ventilated solely through mechanical ventilation, conductance from the air node to the internal surfaces, conductance through the glass surfaces to the exterior, conductance from opaque surfaces to the exterior, and conductance from the internal surfaces to the thermal mass, respectively. The conductance parameters can be calculated using the formulas in Eqs. (B.3)–(B.6), considering the thermal transmittance of the wall U_{wall} , window U_{win} , and the surface areas of the wall, window, floor, and internal surfaces A_{wall} , A_{win} , A_{floor} , and A_{in} . Φ_{HC} represents the heating or cooling power of the HVAC system. Φ_{ia} , Φ_{st} , and Φ_m represent the equivalent heat flows acting on the three temperature nodes T_{air} , T_s , T_m caused by solar heat gain Φ_{sol} and internal heat gain Φ_{int} , respectively. The internal heat flow rates caused by solar and internal heat gains are distributed according to Eqs. (B.7)–(B.9).

$$H_{em} = U_{wall} \cdot A_{wall}, \quad (\text{B.3})$$

$$H_w = U_{win} \cdot A_{win}, \quad (\text{B.4})$$

$$H_{ms} = 9.1 \cdot 2.5 \cdot A_{floor}, \quad (\text{B.5})$$

$$H_{ms} = 3.45 \cdot A_{in}, \quad (\text{B.6})$$

$$\phi_{ia} = 0.5 \cdot \phi_{int}, \quad (\text{B.7})$$

$$\phi_m = \frac{2.5 A_{floor}}{A_{in}} (0.5 \phi_{int} + \phi_{sol}), \quad (\text{B.8})$$

$$\phi_{st} = \left(1 - \frac{2.5 A_{floor}}{A_t} - \frac{H_w}{9.1 A_{in}} \right) (0.5 \phi_{int} + \phi_{sol}). \quad (\text{B.9})$$

In addition, Φ_c and Φ_{air} represent the heat power flowing through the thermal mass heat capacity C_m and the air heat capacity C_{air} , respectively, given by:

$$\phi_c = C_m \frac{dT_m}{dt}, \quad (\text{B.10})$$

$$\phi_{air} = C_{air} \frac{dT_{air}}{dt}. \quad (\text{B.11})$$

The Eqs. (B.10) and (B.11) can be discretized using the Crank-Nicolson method with a given time step Δt as Eq. (B.12). The average values of the node temperatures T_{air} and T_s over the time step can be calculated using Eq. B.13:

$$\phi_i = C_i \frac{T_i^t - T_i^{t-1}}{\Delta t}, \quad (\text{B.12})$$

$$T_i = \frac{T_i^t + T_i^{t-1}}{2}. \quad (\text{B.13})$$

In Eqs. (B.12) and (B.13), C_i represents C_m or C_{air} , and T_i represents T_m or T_{air} .

Using these equations, the indoor temperature T_{air} can be solved numerically.

Appendix C. Validation of the 5R2C model

The 5R2C thermal model is evaluated using SHASE guidelines [54] based on ANSI/ASHRAE standard [55]. A single-zone building located in Denver, USA, served as the test building, as shown in Fig. C.18. The specific test cases 600FF (without nighttime ventilation) and 650FF

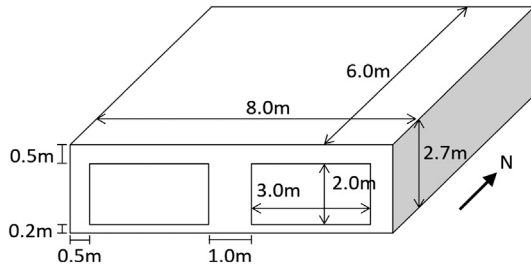
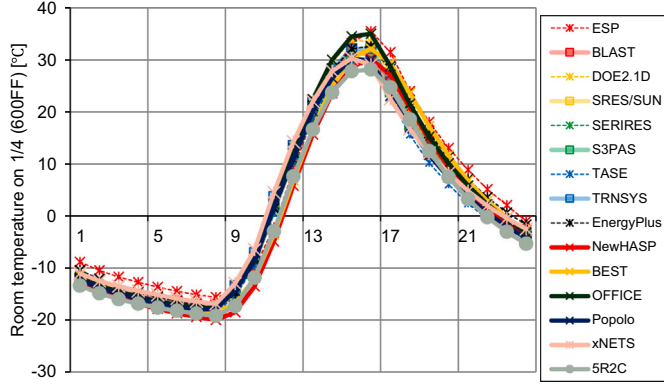
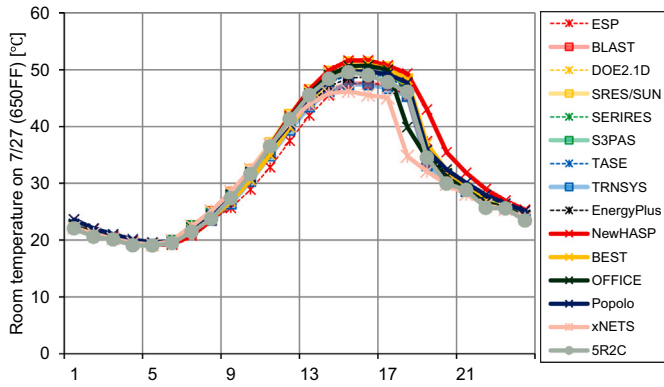


Fig. C.18. Geometry of the test building.

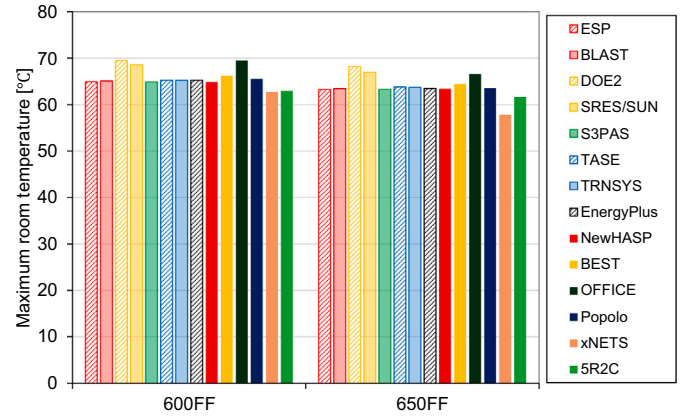


(a) Case 600FF.

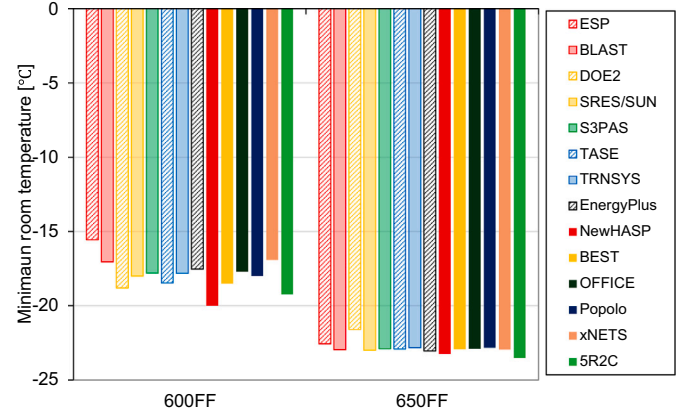


(b) Case 650FF.

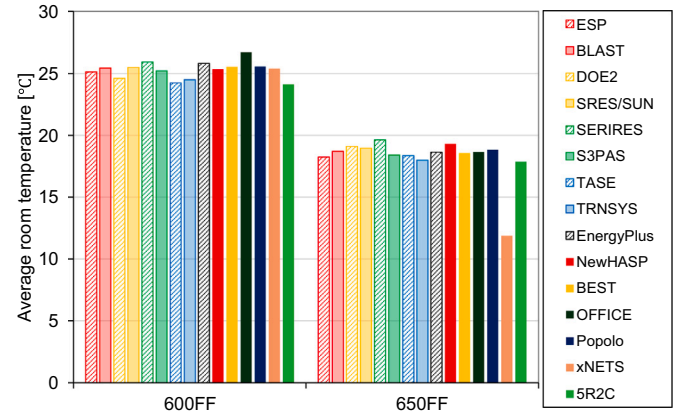
Fig. C.19. Annual maximum, minimum, and average natural room temperatures among different models.



(a) Maximum natural room temperatures.



(b) Minimum natural room temperatures.



(c) Average natural room temperatures.

Fig. C.20. Natural room temperature variations among different models.

(with nighttime ventilation from 18:00 to 7:00) are used to simulate natural indoor temperature. Fig. C.20 compares natural indoor temperature variations among different models, while Fig. C.19 compares annual maximum, minimum, and average temperatures. The results of our model exhibit high consistency with those of other models, demonstrating its reliability in representing building thermal dynamics.

Data availability

Data will be made available on request.

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